

Update Day 3 – June 24th 2021

Onsite groups (RUPPERT gebouw)

Jeroen - RUPPERT – blauw

Opening	11:00
Bergmans	11:15
Driessen	11:30
Guldenaar	11:45
Posterpitches 1	12:00
Poster session 1 & (lunch)break (1.25 hours)	12:15
Nuesink	13:30
Anastasopoulou	13:45
Prummer	14:00
Pruthi Kola	14:15
Posterpitches 2	14:30
Poster session 2 & break (1 hour)	14:45
Schouten	15:45
Makarov	16:00
Closing session	16:15

Stella - RUPPERT – 042

Opening	11:00
Beumer	11:15
Bogert	11:30
Kalogeras	11:45
Posterpitches 1	12:00
Poster session 1 & (lunch)break (1.25 hours)	12:15
Simons	13:30
Teunisse	13:45
Tse	14:00
van Zijderveld	14:15
Posterpitches 2	14:30
Poster session 2 & break (1 hour)	14:45
XIE	15:45
Mutis	16:00
Closing session	16:15

Poster presentations group Jeroen:

Session 1: Soraya Bresser, Simon Buil, Sonia Fanlo Garcia, John Kieran Mc Donagh

Session 2: Bente Klein Hazebroek, Maaïke Klomp, Kimberley Kuik, Alistair Weir

Poster presentations group Stella:

Session 1: Ingeborg Haers, Kevin Hardeman, Robin Heemskerk, Demi Jansen, Ambroos Brouwer

Session 2: Emma Maasland, Alieke Struik, Yannick van Dijk, Shianne van Mierlo, Cato Zantman

Online (MS Teams, Academic Professional)

Christoph – online, MS Teams

Opening	11:00
Deng	11:15
Mohammad	11:30
van der Gouwe (video)	11:45
van Leeuwenkamp	12:00
Lunchbreak (30 mins)	12:15
Woods	12:45
Campisano	12:55
Vossen	13:05
Sieben	13:15
Franssen	13:25
Break (20 mins)	13:40
Gupta	14:00
Simon	14:15
Sheedy (video)	14:30
Pelsma	14:45
Closing session	15:00

Group 1 (Jeroen, RUPPERT - blauw)

Name Student: Anne Bergmans

Internship or Thesis: Thesis

Title project: Predicting cognitive resilience to sleep deprivation through ANS responsivity to social stressors.

In a broad range of professions sleep deprivation (SD) is unavoidable. SD is known to negatively affect cognitive functions which are important for correctly and safely executing certain jobs. However, there are substantial individual differences in how severely one's cognitive functioning is affected by SD (cognitive resilience). Predicting such cognitive resilience would be of great value in certain industries, however underlying mechanisms remain undefined to date. The current study investigated the predictive properties of ANS responses to a laboratory social stressor (Sing-a-Song Stress Test) for cognitive resilience to SD. Furthermore, the widely used Trier Social Stress Test (TSST) and the novel Sing-a-Song Stress Test (SSST) were compared on capacity to elicit similar ANS responses. 89 participants (40 control group, 49 SD group) participated in a 2 day (and 1 night for the SD group) study where cognitive functioning and ANS responses to social stress tests were measured before and after the night of no/normal sleep. Vigilant attention was measured using the PVT and executive functions using the Go/No-Go task, Flexibility task and Sternberg WM task. ANS responsivity to a stressor was measured through HR, EDA, positive affect and negative affect responses. Unfortunately, there were no strong indications that SD induced cognitive decline could be predicted through ANS responsivity. The two stress tests triggered fairly similar ANS responses, except for the SSST generating a stronger HR response.

Name company or University: TNO Soesterberg

(If applicable) External supervisor: Charelle Bottenheft and Anne-Marie Brouwer

UU Supervisor: Jeroen Benjamins

Name Student: Esti Driessen

Internship or Thesis: Thesis

Title project: The MADS app: Is it user friendly?

Obesity in young people is a growing problem. Yet, there are few suitable interventions for this age group. That is why Ijsfontein designed the MADS app. This app helps young people to learn new behaviors. In this study the usability of the MADS app is examined by means of a pilot study. 21 overweight young people will use the app for approximately eight weeks. To determine the usability of the app the comprehensibility and the user experience are measured by means of surveys and interviews. The results show that there is need for an app that supports overweight young people. Still, there seems to be room for improvements. Firstly, young people miss the game element in the app and are therefore less motivated to use the app. In addition, it takes too long for the young people to get started with learning new behavior. Finally, it turned out to be important that notifications are sent regularly. In short, with some adjustments, the MADS app has the potential to support overweight young people.

Name company or University: Ijsfontein

(If applicable) **External supervisor:** Annelies Wisse
UU Supervisor: Caroline Junge

Name Student: Nadia Guldenaar

Internship or Thesis: Thesis (prolonged to 2021-2022)

Title project: Can you feel it? The experience of emotions in Bulimia Nervosa.

Objective. Bulimia Nervosa is a severe eating disorder that often remains present despite evidence-based treatment. Interventions may not address underlying factors well enough, such as emotion regulation. Bulimia Nervosa patients report difficulties in identifying and describing their own emotions, which makes regulating them especially difficult. Emotions include different systems, including cognition, physiology and behaviour. Aside the often-studied cognitive domain of emotion regulation, little is known about a more objective domain: The physiology of emotions. Hence, this study will analyse the physiology of experiencing emotions. **Methodology.** An experiment with a repeated-measures design was created in which both cognitive and physiological experiences of emotions are measured. This study will include both healthy controls and Bulimia Nervosa patients, enabling the comparison of the two groups. By comparing the Bulimia Nervosa group to the healthy controls, we will analyse whether the two groups differ in (1) cognitive emotion experience, (2) physiological emotion experience, and (3) whether the concordance between the cognitive and physiological emotion experience differs.

Name company / University: Utrecht University & Rintveld Center for Eating Disorders, Altrecht

External supervisor: Unna Danner

UU Supervisor: Yentl de Kloe

Name Student: Satyana Nuesink

Internship or Thesis: Thesis

Title project: The Need for Employee Self-management and Work Outcomes

In today's technology, service, and information-driven world, organization increasingly decentralize leader tasks as it is impossible for one leader to possess all the abilities to lead knowledge work. As a result, the need for employee self-management (NESM) within organizations rises. This study set out to examine the effects of NESM on work outcomes and mechanisms that can explain and influence these effects. A cross-sectional survey study was conducted amongst 147 Dutch adult employees. Several hierarchical regression, mediation, moderation, and moderated mediation models were tested in SPSS. Results showed that NESM positively relates to extra-role performance and work engagement, but not in-role performance. The relationship between NESM and extra-role performance was stronger for employees with higher levels of self-efficacy. The moderation effects of personal initiative and empowering leadership were non-significant. Furthermore, authenticity (but not intrinsic motivation) mediated the relationship between NESM and extra-role performance (partial mediation) and in-role performance (full mediation). No moderated mediation effects were found. In conclusion, results implicate that employers can benefit from the decentralization of leader tasks. It forces employees to exert self-

management which positively affects authenticity, work engagement, and extra-role work performance.

Name company or University: Utrecht University

UU Supervisor: Toon Taris

Name Student: Anastasopoulou Theofania

Internship or Thesis: Thesis

Title project: Getting feedback on engaging educational concepts for people with Chronic Obstructive Pulmonary Disease (COPD) - an online study

Chronic obstructive pulmonary disease (COPD) is a progressive and irreversible disease characterized by respiratory symptoms such as dyspnea and cough with excessive mucus production. Educating COPD patients on their disease, is important for their self-management as they can take better decisions about their health. However, current educational programs offered to patients are one-size-fits-all, resulting in unclear picture of their disease, therefore more hospitalizations. Based on constructs taken from Self Determination Theory, Autonomy, Competence and Relatedness, the goal of this study is to explore how education for COPD patients can become more engaging and relevant. To do so, we designed mock-ups of an educational module that incorporated the mentioned three constructs in three scenarios. To evaluate whether they are conveyed sufficiently, an online survey will be conducted. The participants will be presented with the mock-ups and questions for each scenario. The results will yield important insight that will potentially inform the development of education in the context of a future self-management application.

Name company or University: Philips

(If applicable) External supervisor: Saini Privender

UU Supervisor: Niilo Valtakari

Name Student: Franziska Prummer

Internship or Thesis: Thesis

Title project: The Influence of Robotic Tasks on Collaborative Behavior of Children with ASD

Autism Spectrum Disorder (ASD) is mainly characterized by a deficit in social skills (Weiss & Harris, 2001). Children affected by this neurobiological disorder generally experience difficulties with social communication, establishing peer relationships, initiating, and participating successfully in reciprocal interactions (Goldstein, 2002; Lord et al., 2000, 2018; Wing, 1997, 1989). Studies have proven a relationship between the motor skills of children with ASD and their social skills (MacDonald, Lord & Ulrich, 2013; 2014). As a result of the impairment of their social skills, it is reasonable that the collaborative skills of children with ASD are typically at a lower level compared to children without the disorder. The aim of this study was to investigate the influence of robotic tasks and robot complexity on collaborative behavior of children with ASD. Two different robot types differing in complexity of appearance and use were utilized and children in four different schools and educational institutions participated in group robotic activities. They were given the task to create and complete a parkour with the robots and to collaborate with each other. Video recordings supported by coding schemes and questionnaires directed at teachers

measured and assessed collaborative behavior of children. Predicated on prior research, it was hypothesized that the complex robot task would elicit more collaborative behaviors between the children, as this robot induces more practice of motor skills. **No results yet, as data is still being collected.*

Name company or University: Utrecht University & Robowijs

External supervisor: Annika Hellendoorn

UU Supervisor: Christoph Strauch

Name Student: Sara Pruthi Kola

Internship or Thesis: Internship

Title project: Predicting cognitive failures from preceding brain activity using EEG

The aim of the present study was to investigate whether we can predict cognitive failures from preceding brain activity using frontal fP3 EEG responses. The experiment consisted of a visual task in which the participants monitored a continuous stream of patterned stimuli centrally presented and flickering at a rate of 12.5 Hz. The participants were required to press a response key (spacebar) as quickly as possible when they detected a frame of longer duration (target). The data were preprocessed using the Brain Vision Analyzer 2.1 program. As hypothesized, fP3 amplitude was increased before a hit (target detected) relative to a miss (target not detected). Based on the results, fP3 seems to be a good predictor of susceptibility.

Name company or University: Utrecht University

UU Supervisor: Leon Kenemans

Name Student: Lisanne Schouten

Internship or Thesis: Thesis

Title project: Does the pill impede unlearning fear?

Despite the fact that the contraceptive pill is used by over 100 million women around the world, research about psychological side-effects is limited. There are strong indications that hormonal alterations due to OC-use impact on socio-emotional behaviour and underlying brain functions. In particular, research has demonstrated negative effects for fear regulation processes. However, a lot is still unclear and only small effects have been found, partly due to poor experiment design. It is important to clarify the exact impact of the pill for fear regulation in order to properly educate women about this. Aim of this study is to contribute to such a clearer understanding, by investigating the role of OC use in a fear extinction experiment. This study investigates whether women using the pill struggle extinguishing fear as compared to non-users. This will be done by comparing subjective fear scores that are provided by participants at different times during a fear conditioning and fear extinction task. As the experiment will be conducted online, hopefully a large sample can be gathered that includes subgroups which are often underrepresented in related lab studies. Here for instance, we hope to gather a large enough group of naturally cycling women to use as a control group. By knowing whether the pill really impairs fear extinction processes not only education can be improved, but hopefully also fear and anxiety treatments can be optimised

for the suffering group, and potentially prevented for women who may run a higher risk of mood and anxiety side effects of the pill. Additionally, an effect of pill phase will be examined. This study investigates whether pill users in an active pill week perform worse in extinguishing fear as compared to pill users in a pill-free week. I expect that OC-users will end up with higher fear scores after the online extinction task as compared to non-users, and that they will be overrepresented in groups of poor fear extinction. Furthermore, it is hypothesised that OC-users in an active pill week will end up with higher fear scores after extinction training as compared to pill users in a pill-free week, and that they will be overrepresented in groups of poor fear extinction as well.

Name company or University: Utrecht University / Department of experimental psychology

(If applicable) External supervisor:

UU Supervisor: Prof. Dr. Joke Baas

Name Student: Ivan Makarov

Internship or Thesis: Thesis

Title project: Exploring of access and cost of storage in the homeostasis model of the VWM using a visual search paradigm

Our brain can store only a limited amount of visual information in working memory. For now, there are no theories that could explain how the brain balances between storing information in memory and using world around us as an external memory storage. Several studies describe how eye movements help to keep the balance between the external world and the internal working memory. The current study investigated how a physically presented target template as an example of external memory storage would influence traditional visual search task paradigm. The complexity of a task was used to overload visual working memory. During hard trials participants made more saccades from a target to template than on the easy trials. Current results showed that the participants prefer to rely on the physical template regardless the distance to it. Our findings reveal that it is less expensive to make the saccades to compare a template compared to storing it in visual working memory.

Name company or University: Utrecht University

(If applicable) External supervisor: no

UU Supervisor: Stefan van der Stigchel

Poster pitches 1 & Poster session 1

Name Student: Soraya Bresser

Internship or Thesis: Internship

Title project: Static vs. dynamic light: Comparing the effects of dynamic compared to single colored light on relaxation in the multipurpose space Megg.

The Megg is a multipurpose space which is intended to allow people to detach, relax and recharge. This space is designed in a way that users can close the door and spend some time doing what they want, without getting disturbed. The space will be provided with

various technologies suggested to improve human well-being. One of these technologies is lighting. Light is important for the sleep-wake cycle, health and well-being. This research aimed to test what the difference is between single colored lighting compared to dynamically changing colored lighting on the experience of relaxation and regained alertness. A sample of 30 participants experienced 2 sessions of 5 minutes in the Megg. Five minutes of static green light and 5 minutes of dynamically changing light changing from green to blue after 4 minutes. Five images of the SAM rating scales were used to measure arousal and valence. Five-point scales were used to measure maximum relaxation and alertness. The static condition was perceived as equally arousing, as well as equally pleasant, compared to the dynamic condition. Additionally, the two conditions were not found to differ in the experienced maximum relaxation during, as well as alertness after, the session in the Megg. Although participants selected the static condition more often as being more relaxing. And finally, participants did not prefer one condition over the other for a within-day work break. Majority of the participants mentioned to prefer to decide on a color upon entering the Megg, based on their mood and the desired goal of using the Megg. Based on current findings it was recommend to Megg to give participants multiple options for colored lighting, taking in mind different color-preferences and desired goals upon entering the Megg.

Name company or University: Megg World

(If applicable) External supervisor: Suzanne Plesman

UU Supervisor: Christoph Strauch

Name Student: Simon Buil

Internship or Thesis: Internship

Title project: The flaws in AGL-studies

My internship was about the way in which animals can learn syntax like rules. They developed artificial grammars to test these abilities, but there is one flaw present in many of the AGL-studies. The flaw is that there are more acoustic similarities in correct strings then there are similarities in the incorrect strings. During my internship I tested 15 AGL-studies on their amount of acoustic similarities divided over 4 categories (contains, shared start, shared substring and cross-correlation). I found that only 2 out of the 15 passed the test of acoustic similarities and we can thus conclude that future AGL-studies should focus on overcoming the flaw of acoustic similarities and that the moment we can not conclude much over the grammar ability of animals.

Name company or University: Utrecht University

UU Supervisor: Gabriel Beckers

Name Student: Sònia Fanlo Garcia

Internship or Thesis: Internship

Title project: Multipurpose space: UX study and insights

Abstract: There is an increasing amount of evidence humans need several breaks during the day. With the increase of corporate culture the office became the main space individuals spent time at, and after everyone's experience working from home and juggling both work and house responsibilities parallelly, there is an even higher awareness about the benefits of taking a small period of time to disconnect. Megg is designed as a multipurpose space

promoting relaxation, privacy, sustainable human performance (work), personal energy management, videocalls and meditation. Because Megg is still not on the market, in this study we want to study 1) What are the potential users' expectations in regards to the product fulfillment functions? And 2) What features would potential users like to have in the space that would help them obtain the desired experience. Participants tested the product for 10 minutes and answered a questionnaire about their experience and gave their input on their suggestions for the next prototype features.

Name company or University: Megg World
(If applicable) External supervisor: Suzanne Plesman
UU Supervisor: Krista Overvliet

Name Student: John Kieran Mc Donagh
Internship or Thesis: Internship
Title project: Investigating the ecological validity of military relevant stimuli in a decision making task.

Researchers studying soldiers' cognitive performance have, up until recently, mainly conducted such studies using experiments that are not relevant to military contexts. As a consequence of the lack of ecological validity in these experiments, a demand has arisen for cognitive tasks that more closely simulate aspects of real world military activities. In the present study, an attempt was made to modify an existing 'expanded judgment' task to increase its ecological validity within military contexts. Two new tasks were developed through adjustments that made use of military relevant stimuli (muzzle flashes from firearms instead of flashing circles), and performance on these tasks was subsequently compared. The aim of the study was to calibrate and evaluate the reliability of the newly generated tasks, as a preliminary test before military assessment. Thirty-two students (16 male, 16 female; $M = 22.9$, $SD = 4.4$) participated in the study. A one way ANOVA indicated that there was a highly significant change in sensitivity between the tasks, $F(2, 87) = 13.797$, $p < .001$. Post Hoc Tukey tests revealed that there was a significant difference between high ($M = 1.2911$, $SE = .079$) and medium ($M = 1.643$, $SE = .088$) ecological validity, and low ($M = 1.935$, $SE = .$) and high ecological validity. Performance was worst in the condition with the highest ecological validity. Whilst it is incorrect to generalize this to all military tasks and assume that it is better to have military relevant stimuli. We argue that in this specific context, having stimuli with higher threats that induce higher levels of arousal are more beneficial as they are more representative of military tasks.

Name of company: TNO
(If applicable) External supervisor: Dr. Maurice van Beurden
UU Supervisor: Dr. Leendert van Maanen

Poster pitches 2 & Poster session 2

Name student: Bente Klein Hazebroek
Internship or thesis: Internship
Title project: Visual Expertise among Industrial Designers and Architects

Industrial designers and architects mainly work in the visual domain. They can create and imagine visual ideas, and are trained in visualising complex properties of the environment. Their expertise has to fit the demands of today's dynamic society, but it is not yet clear what cognitive skills and processes are crucial to perform such visual tasks, and how they can best be taught. This study explored what design teachers believe is important in the development of visual expertise of industrial designers and architects, and how visual expertise is addressed in academic design programmes. By looking at the content of European academic design programmes, and interviewing experienced teachers in design, the aim of the internship was to make an overview of factors that play a role in the development of visual expertise among industrial designers and architects. Teachers were asked what they believe is important for students' visual expertise, and how this is taught in the programme(s) they teach. They report that novice designers still have to develop skills like spatial imagination, visual awareness, and visual communication, lack the knowledge of design elements that they can work with, and need to refine their ability to apply this knowledge with different tools and methods. Programmed instruction, experiential learning, and interactive classes are methods that were mentioned to teach visual expertise. Many skills related to designers' visual expertise take time and experience to develop.

Institute: Utrecht University

UU supervisor: dr. prof. Susan te Pas

Name Student: Maaïke Klomp

Internship or Thesis: Internship

Title project: Supreme Nudge app – facilitating healthy lifestyle behaviours of low-SES adults with a mobile application Supreme Nudge is a research project that is concerned with sustainable prevention of cardiometabolic risk through nudging health behaviours for low-SES adults. The development and testing of a mobile application that encourages people to do more physical activity is part of this project. The application is able to provide personalised time- and context-specific feedback to the user. In my internship I assisted with a pilot-test of the mobile application. The aim of this internship was threefold: 1) creating questionnaires from literature, 2) conducting and transcribing interviews for the qualitative part of the pilot-test and 3) developing personalized physical activity messages based on the weather for the mobile coach within the app. By doing this, information was gained on how the mobile application could be improved for the main research study and the message database used in the study was expanded.

Name company or University: Amsterdam University

External supervisor: Gert-Jan de Bruijn

UU Supervisor: Caroline Junge

Name student: Kimberley Kuik

Internship or Thesis: Internship

Title project: The Relationship Between Intolerance of Uncertainty and Fear Extinction and Generalization

For my internship, the relationship between the characteristic intolerance of uncertainty, and fear extinction and generalization has been investigated. Research has shown that the extent to which people get rid of fear or generalize it depends on several individual differences between people, such as various personality characteristics. One of these is intolerance of uncertainty, which means that someone is disturbed by the fact that something negative might happen, even though the chances of that event occurring are small. Based on previous findings, the expectation was that people who show generalization of fear and poor fear extinction have a higher score on the intolerance of uncertainty scale. To test whether this was true, data from an online pilot version of the experiment was used. Intolerance of uncertainty was measured by the Intolerance of Uncertainty Scale (IUS), which consists of 27 items describing a certain situation, for which the participants need to rate how they would feel in that situation. For the fear extinction and generalization, participants were divided into groups based on their score in the experiment. Against expectation, no significant differences on IUS score were found between different groups, meaning there is no relationship between intolerance of uncertainty and fear extinction and generalization. More research is needed to get a better view on this relationship.

Name company or university: Utrecht University
UU supervisor: Joke Baas

Name Student: Alistair Weir
Internship or Thesis: Thesis
Title project: Does Increased Cognitive Load Impact Response Times to Your Name?

This present study investigates how visual cognitive load affects participant's response times to hearing their name. I draw on previous work that found when subjects were visually distracted, they were faster in responding when their name is included in the alert. I created a dual task that asks subjects to respond to auditory alerts (by pressing either pressing a left or right key) that are preceded by either their name or a random name, while they undergo a multiple object tracking task. At the end of the task, a single object was highlighted and participants stated whether or not it was a focus object. The multiple object tracking task increased in difficulty and performance for both tasks was recorded. This enables us to observe how visual attention affects response times to auditory alerts. Response times were faster when the alert was preceded by their personal name, even as the visual load increased. Data suggests we still attune to our name while sustaining visual attention.

Name company or University: Utrecht University
UU Supervisor: Dr Christoph Strauch

Group 2 (Stella, RUPPERT - 042)

Name Student: Teun Beumer

Internship or Thesis: thesis

Title project: Investigating the relationship between spatial frequency and our visual representation of the world

The ostensibly easy question of how humans perceive the world around them has been the topic of investigation for decennia. Odd enough, when the general gist of a scene stays roughly the same, people are oblivious for small changes in detail. Furthermore, people have no trouble reporting the mean emotion of a set of faces without being able to report the emotion of one of the individuals. All this points to people processing visual cues top-down. This, however, gives no insights in how the elaborate span of information entering via the visual system is coded in the brain. The hypothesis of this study is that spatial frequency is one of the components with which the brain constructs daily reality. In a two-alternative-choice task, 49 participants were asked to distinguish a happy display from an angry display. These displays also showed the spatial frequency of happy and angry faces. The expectation was that if a display showed angry faces and angry spatial frequency noise the participant wouldn't need to see as many angry faces to discern a display as angry as opposed to when a display included angry faces with the spatial frequency of happy faces. Analysis showed that the amount of angry faces needed to discern a display as angry did not differ based on the spatial frequency. Therefore the hypotheses was rejected.

Name company or University: Utrecht University

UU Supervisor: dr. Sjoerd Stuit

Name Student: Dorien Bogert

Internship or Thesis: Thesis

Title project: Developing serious games: The impact of narrative on the transfer of learning goals in *Ruby's Mission*.

Abstract: "Many studies have shown that children with chronic conditions are at a greater risk for poor mental health and social problems" (Utrecht University, n.d.). With this in mind, Utrecht University initiated the *Applied Gaming Intervention for reducing Loneliness of Children with Chronic Diseases*. The game for this gaming intervention is called *Ruby's Mission*. This videogame is developed to increase the players' emotional awareness through reaching certain learning goals. For the current research, a narrative was developed, based on literature. Through performing an experiment, it was tested if inclusion of this narrative caused a difference in transfer of the learning goals, compared to the non-narrative group. Based on the literature, we hypothesized that the narrative condition should increase their emotional awareness even more than the non-narrative group after a gaming session. The results are yet to be analyzed. For now, it seems like the questions used to gauge emotional awareness were more difficult after the session than before, which caused an overall negative trend of emotional awareness between before and after the session. Moreover, it does seem like the inclusion of a narrative did not have a significant impact on this trend.

Name company or University: Utrecht University
External supervisor: Dionysis Alexandridis (game research)
UU Supervisor: Zsolia Bélteki

Name Student: Dimitrios Kalogeras
Internship or Thesis: Thesis
Title project: Cognitive Performance During Spatial Disorientation

Conflicting motion cues can cause spatial disorientation (SD). In aviation, SD is a common hazard in human error mishaps, since it can affect a pilot directly (inappropriate control inputs, startle), or indirectly by competing for cognitive resources with concurrent tasks. "Orientation first" principle explains that resolving SD can be prioritized to other tasks, like a reflex. Moreover, the notion of surprise can be an integral part of SD, where the reappraisal of a situation requires cognitive resources. Reorienting oneself in an aircraft may therefore draw attention away from other cognitive tasks. This study aimed to distinguish the effects of motion magnitude (high/low) and SD (yes/no) on cognitive performance, as decrements in a counting task. Post-rotary effects were utilized to manipulate incongruent vestibular and visual information. The findings revealed a significant effect of SD in the number of counts, yet no significant effect in errors. Motion magnitude, and its interaction with SD, showed no significant effects.

Keywords: spatial disorientation, motion magnitude, cognitive performance, surprise

Name company or University: Utrecht University/ TNO
External supervisor: Annemarie Landman
UU Supervisor: Baptist Lifooge

Name Student: Rogier P.J. Simons
Internship or Thesis: Thesis
Internship/thesis Title project: What am I looking at? - Towards Automatic Video Annotation for Mobile Eye Tracking Research
The development of wearable eye trackers in Eye Tracking research allowed for more freedom in the experimental setup, but at the cost of extensive manual analysis time. This research tried to decrease manual analysis time of mobile eye tracking experiments by developing a technique for semi-automatic annotation of the video material. First, different approaches proposed by other research were analysed, to find possible improvements in the current solutions. Using object detection models seemed most promising, but a big problem with this technique are the extensive training sets that are required to train these models. This research proposes a new way to annotate a small part of the video material, which can be used to create the required training sets with less manual effort than in traditional annotation tools. With this new proposed approach an object detection model was trained that was able to label around 70% of the Areas of Interest in the video material. This new approach becomes feasible after multiple participants, when the initial time it takes to gather the training data weighs up to the manual annotation of the gaze

fixations, but from that moment this approach seems quicker than other proposed solutions for this problem, with the added freedom of training on custom objects.

Name company or University: Utrecht University

UU Supervisor: Ignace Hooge

Name Student: Ava Tse

Internship or Thesis: Thesis

Title project: Are you looking at the most salient area first on web pages? Comparing mouse and eye movements with saliency map

An understanding of how people allocate their visual attention when viewing webpages is very important for web authors, interface designers, advertisers and others. Eye-tracking is one of the most widely used tools to understand viewing behavior over webpages. However, collecting gaze data with traditional eye-tracking is expensive, tedious and slow which makes web creators hard to benefit from the insight of eye movement. Visual attention modeling has been an area of active research over the past quarter of century. AlphaOne, a neuro research company has successfully deployed a neural network on predicting saliency map on the natural image but the effectiveness of the neural network with context rich scenes like webpages are still unknown. The high level and long-term goal of AlphaOne is to train a neural network predicting scanpath on webpages. With the benefits of crowdsourcing platforms like Amazon Mechanical Turk, mouse movement experiment is more scalable comparing with traditional eye movement experiment. Thus mouse movement data is more preferable for cost-effective and large scale training on the neural network. However, the studies on viewing pattern comparison of mouse and eye movements and scanpath on multiple users on web pages are limited. Base on the study on 51 webpages with eye and mouse movement data from 25 subject who free-view the webpages, this thesis attempted to validate mouse movement heatmaps and computer vision saliency map to eye movement heatmaps, determined the entry point within datasets and proposed a methodology producing a general scan path at multiple users level.

Name company or University: AlphaOne

(If applicable) External supervisor: Coert van Gemeren & Ingrid Nieuwenhuis

UU Supervisor: Samson Chota

Name Student: Remco van Zijderveld

Internship or Thesis: thesis

Title project: BioFace: a feasibility pilot of EMG biofeedback training on facial expressions to regulate stress.

Stress is an adaptive response to physiological and psychological threats, also named stressors. Too much-prolonged stress can be problematic and might result in pathologies, which makes it crucial to prevent chronic stress. This could be done with the help of biofeedback, which is a widely used method to make one's aware of their physiological responses. Although a lot of research is conducted on heart rate variability biofeedback,

little is known about biofeedback based on muscle tension. Therefore, this study aimed to explore the feasibility of biofeedback based on facial expressions. 16 healthy individuals underwent biofeedback training and performed a stress task twice. The stress task was performed before and after the training session. During the second assessment, subjects received biofeedback if their facial expression was too tensed. Facial muscle tension was obtained through EMG electrodes. As a subjective measure of stress, subjects were asked how stressed they felt, whereas heart rate variability served as an objective measure for stress. Because the data acquisition was just completed, the analysis has not been finished yet. Therefore, results and possible conclusions are still missing.

Name Company or University: Military Mental Health Care/ Dutch Ministry of Defense

External supervisor: Antoin de Weijer

UU Supervisor: Ruud Hortensius

Name Student: Peiqi Xie

Internship or Thesis: thesis

Title project: Happy or Neutral: Face Masks Make It Harder to Read People's Emotions

COVID-19 has been one of the most devastating pandemics in history. The World Health Organization has suggested wearing face masks in public places to slow the spread of the virus, as this is a critical step to prevent infection and save lives. However, as more people cover their faces in public, reading facial expressions and seeing people happy or angry has become more difficult. We investigated the effect of face masks on the readability of emotions in this study, which included two online tasks. The participants (N = 99) were asked to choose which of the seven labels (neutral, surprise, fear, sad, disgust, angry, and happy) best described the facial expression in the naming task. In the rating task, the participants (N=99) were asked to rate the facial expressions on three dimensions by the Self-Assessment Manikin (SAM), which included valence (positive / negative), arousal (excited / calm), and control (dominant / submissive). The within-subjects design was used for both tasks, which included face gender (male vs. female), facial expressions (seven: neutral, surprised, afraid, sad, disgusted, angry, and happy), and the face mask (with vs. without a face mask). The blue surgical masks were used to cover the facial expression stimuli, which were selected from the Karolinska Directed Emotional Faces. JASP will be used to analyze the data.

Name company or University: Utrecht University

UU Supervisor: Dr. Martijn Mulder

Name student: Yarno Felix Mutis

Internship

Are people able to fight and detect Deepfakes by themselves? A conflict detection study.

Abstract: The internet is becoming a chaotic environment, saturated with mis- and dis-information which in turn confuses the public about certain (important) topics. FakeNews

articles contribute to creating a confusing information sphere and because of technological advancements and a focus on AI development in the last couple of years it is now also possible to confuse people with so-called *Deepfake* videos. In these videos the AI superimposes someone's facial features on-top of another or the same face. The technology has advanced so much that the videos look and feel very authentic, and because authentic recordings afford credibility and uphold epistemic values Deepfakes are a great medium to spread FakeNews. Deepfake detection software is created to combat this technology however it will probably fall short in many cases as it 1) comes after the video is already spread online, 2) the results might not reach the whole population, 3) the results might not be believed by everybody. In this study we are therefore interested to discover if humans are able to distinguish Deepfake videos from authentic videos. Furthermore we are interested to see signs of conflict detection when people believe the Deepfake is authentic. Heuristics-and-biases studies show how people doubt their answer more when they choose the incorrect heuristic option in a reasoning task, meaning they are somewhat aware of their biased reasoning. Believing in a Deepfake is similar to choosing the heuristic option in a reasoning task and consequently we saw how participants were significantly less confident about when they believed in a Deepfake then when they believed in an authentic video. Moreover participants were able to detect Deepfakes but only to a certain extent, and a warning about the presence of Deepfakes and what they are increases their accuracy but also makes participants believe authentic videos are Deepfakes more.

Name Company or University: University of Utrecht

External supervisor: Eva Janssen, PhD

UU Supervisor: Dr. Stella Donker

Poster pitches 1 & Poster session 1

Name Student: Robin Heemskerk

Internship or Thesis: Thesis

Title project: Managing anger in Autism Spectrum Disorder (ASD): a cerebellar neuroscience approach

This study looks at the relationship between cerebellar grey matter volumes and aggression in children with ASD and neurotypical (NT) children. Understanding the neurological processes underpinning the elevated aggression rate in ASD may be the key to more effective treatment.

Previous research shows indications of the cerebellum's involvement in aggression, specifically lobules right Crus I and II and the vermis.

Volumetric data from 200 3T MRI scans of children with ASD and NT children, ages 8-13, was used. The aggression measures were extracted from the CBCL.

Linear regression analysis showed a significant relationship between the grey matter volume of right lobule VIIIA and aggression scores in children with ASD.

Lobule VIIIA is located in the right posterior area of the cerebellum. The cerebellum is contralaterally associated with the frontal cortex, where aggression correlates with relatively higher activity on the left. This right lateralization is strongest in the posterior lobe.

Aggression in children with ASD may stem from impaired somatosensory processing, since overstimulation can lead to aggression. This may explain the implication of the cerebellum, since it plays a role in somatosensory processing.

Name company or University: Utrecht University
UU Supervisor: Dennis Schutter

Name Student: Ambroos Brouwer

Internship or Thesis: thesis

Title project: The effects of priming on the use of scene guidance to improve search performance for military targets in naturalistic outdoor scenes.

One of the most important factors during the conduction of military operations is to stay undetected. Therefore, the military could benefit greatly from a model that can predict the detectability of their assets. Unfortunately, previous eye-movement models only incorporate target salience, while disregarding important top-down cognitive processes, such as scene guidance. In scene guidance, people use learnt target-context associations to interpret scenes and to make predictions about potential target locations. In the current experiment, it is investigated whether such target-context associations can be learnt through priming and whether search performance was higher for primed participants than for their non-primed counterparts during search for military targets in naturalistic heathland scenes. The results show that the priming group did not perform better than the non-priming group. This suggests either that the participants did not learn the target-context association, or that the priming group simply did not develop a more effective search strategy than the non-priming group

Name company or University: Utrecht University
UU Supervisor: Roy Hessels

Poster pitches 2 & Poster session 2

Name Student: Emma Maasland

Name Internship or Thesis: Sleep deprivation research at TNO

Internship/thesis Title project:

Predicting PVT performance from physiological synchrony: an exploration of computational methods and a novel measure of attention

Vigilant attention (VA), the ability to focus for a prolonged period of time while performing a low-level cognitive operation, is heavily influenced by sleep deprivation (SD). VA can be measured using the Psychomotor Vigilance Task. This measurement, however, is obtrusive and time-consuming. We hypothesize that physiological synchrony (PS) in heart rate might

be a way to predict (and therefore, measure) VA passively and unobtrusively. PS is defined as a concurrency in physiological (heart rate, EEG, skin conductance) changes in people (2 or more) attending to the same sensory stimuli for a prolonged period of time. We studied 54 participants during a night of acute SD, while they performed two tasks hourly: watching a ten-minute video (PS elicitation) and performing a ten-minute PVT (VA measurement). We built a hierarchical linear model to see if ISC to group values (PS scores) were predictive for PVT performance. The outcome was not significant, which authors suspect to be due to the fact that PS scores are reflective of different components of attention than PVT performance, which can respond incongruously to acute SD. Additionally, we performed an analysis to discover whether wavelet coherence analysis (WCA) is a suitable method for PS computation when using the chosen stimulus material. WCA was not deemed a fitting computational method for the current study design.

Name company: TNO

External supervisor: Anne-Marie Brouwer

UU Supervisor: Jeroen Benjamins

Name Student: Yannick

Internship or Thesis: Thesis

Title project: Communication over distance: Key features of mediated social touch (MST)

The effects of touch are far-reaching and can be measured throughout the human lifespan. Touch is, however, not always possible, and this is reinforced by the current COVID-19 pandemic. Touch technologies such as mediated social touch (MST) have the potential to compensate for this lack of touch by enabling “physical” interaction. While MST seems promising, results of studies that attempt to mimic the effect of unmediated social touch are mixed. The problem seems to be that research often uses incomplete prototype design. As they focus on certain characteristics of MST, they lack others. We do not know which of these characteristics are important in relation to each other. Therefore, I investigate what characteristics are considered important in MST communication. It could be the case the people’s preferences or needs are not universal. To prepare for this outcome I also exploratively investigate the effect of personality traits on the importance of certain characteristics, so to suggest a possible explanation. A Rasch analysis is used, through data-to-model fit, including rater fit statistics, to test for characteristics importance. This allows me to determine the extent to which the importance of characteristics is indeed universal (or person-free). The effects of personality traits on importance of characteristics will be examined via correlation analysis.

Name company or University: Utrecht University/Eindhoven University of Technology

(If applicable) External supervisor: Antal Haans (TU/e)

UU Supervisor: Anouk Keizer

Name Student: Shianne van Mierlo

Internship or Thesis: Thesis

Title project: Field observations of reactions of incidentally copresent pedestrians to a seemingly autonomous sidewalk delivery vehicle: An exploratory study

Abstract: In the last decade, there has been a rise in the development of autonomous vehicles, including autonomous sidewalk delivery vehicles. However, there are only a few studies that investigate reactions of people to autonomous sidewalk delivery vehicles. Thus, a field observation was done on a sidewalk where a delivery vehicle was driving on a specific trajectory using a wizard-of-oz design. Reactions of incidentally copresent pedestrians to a seemingly autonomous sidewalk delivery vehicle were explored and used to find various themes. The findings from the present study revealed new avenues, concerning pedestrians' natural reactions towards autonomous vehicles, that should be considered to be further investigated for future road safety as well as the design (external appearance and programming) of autonomous sidewalk delivery vehicles.

Name company or University: More Work Less Carbon (MWLC)

External supervisor: Hans Steuten

UU Supervisor: Leendert van Maanen

Name Student: Bas Vossen

Internship or Thesis: Internship/thesis

Title project: The potential of nudging in retaining online customers

This literature review outlines the present state of scholarly research in application of nudging as a tool to retain customers. Literature search is performed by looking for relevant articles through a fixed set of search phrases in three scientific databases. For this review to contribute to an up-to-date view on nudging in an online context, literature is only used if published from 2018 to present date. Post-sale service evidently is an important aspect in retaining customers, enhancing customer satisfaction and achieving a good customer-retailer relationship. Future research should attribute to the gap of research in the application of nudging during the post-purchase stage of the customer journey.

Keywords:

Nudging · Online shopping behavior · Retaining customers · Post-purchase · E-commerce

Name company or University: Utrecht University

(If applicable) External supervisor: -

UU Supervisor: Ignace Hooge

Group 3 (Christoph, online, MS Teams)

Name Student: Na Deng

Internship or Thesis: thesis

Title project: Explore the influence of pitch contour and rate on the attribution of affect to synthetic speech.

The commercialization of synthetic speech through the use of conversational user interfaces is intensifying. As humans, we suppose these voice interfaces to be fully calibrated with our way of social speaking. It is thus expected that a conversational user interface can convey emotions when the situation at hand requests this. However, for now, conversational user interfaces remain incapable of fulfilling these expectations fully. This thesis, therefore, aims to provide a set of recommendations on how synthetic speech could acoustically be manipulated to express a wide range of emotions. The expression of emotion in synthetic voice enables technology to create more natural interactions with its users. A listening experiment presented 5x5 (Pitch contour x Rate) synthetic speech fragments and measured ratings on perceived emotions using the dimensional Self-Assessment Manikin (SAM) and the discrete Product Emotion Measurement Tool (PrEmo).

Name company or University: Eindhoven University of Technology (TU/e)

name (If applicable) External supervisor: Dr Peter Ruijten-Dodoiu

UU Supervisor: Dr. Stella Donker

Name Student: Fatemeh Mohammad

Internship or Thesis: thesis

Title project: Analysis of unsupervised eye tracking data

Goal: The aim of this graduation assignment is to use freely available coding tools to analyze a unique and large dataset collected with unsupervised eye-tracking experiment. Further, the data will allow for explorative analyses into the relationship of central gaze parameters and demographic variables. Data analysis approaches are inspired by different research fields including image processing and machine learning.

Findings: Testing the quality of the data collected in unsupervised eye tracking experiments shows that the quality of the collected data is high enough for scientific studies. Further, looking into the analysis results there is a significant correlation between gender and gaze events. However, this relationship strongly depends on what algorithm has been applied for detecting the events. Another finding relates to the predictability of such data given the demographics. Preliminary data shows that the age can be estimated with an error of around 10 years using neural network algorithms.

Name company or University: Utrecht University

UU Supervisor: Christoph Strauch

Name Student: Tim van der gouwe

Internship or Thesis: Thesis

Title project: Gaze estimation in remote usability research

Usability research has benefited greatly from the rise of the internet. Platforms like usertesting.com and User Sense make it possible to conduct usability tests remotely, which allows for efficient and effective user testing on an (inter)national level. Unlike usability research, gaze estimation studies still take place in a lab. This is due to the fact that appearance-based gaze estimation studies are deemed inaccurate. In this thesis I will discuss the extent to which existing gaze estimation algorithms can be used for remote usability studies and, if so, what that might look like in a real-world setting.

Name company or University: Utrecht University

(If applicable) External supervisor:

UU Supervisor: Ignace Hooge

Name Student: Alexandra van Leeuwenkamp

Internship or Thesis: Thesis

Title project: Is it possible to measure expected well-being?

Goal: In this research, I want to explore 'What is the state of the art concerning asking people to estimate the effect of policies on their experienced wellbeing?'. The goal of this research is to develop and test a new survey that can measure expected well-being, controlling for the hyperbolic discounting bias. **Background:** There is growing discontentment among citizens, academics and policy-makers towards the way most governments define and measure progress, by emphasising economic indicators such as Gross Domestic Product (GDP). It has grown evident that well-being should be measured by governments so as to understand the impact decisions will have on well-being at a national level and should serve as the end goal of national public policy. Although there is a lot of research on how to measure well-being, there is little that looks at measuring expected well-being for policy decisions. **Method:** As there is currently no survey that has been designed to measure expected well-being, we developed our own. The survey is based on the Investigating Choice Experiments Capability measure (ICECAP), developed by the University of Birmingham by Janabi, Flynn, Coast (2012). ICECAP is based on Sen's capability approach, assessing well-being in terms of what people do and are (functioning), and what people are able to do and be (capability) in their lives. We adapted the ICECAP-A measurement to try and capture expected well-being. This was done by changing the wording in the original questionnaire to encourage participants to think of themselves in the future. Inspired by the research on the 'Different Lives Approach' by Adler and Dolan (2008). In which respondents are asked to rank or rate 'Different lives', characterised in terms of different dimensions of overall quality of life. Additionally, research by Tatsuyoshi Saijo and Keishiora Hara from Japan known as 'Future Design', where people were designated to 'take on the role of future generations'. These individuals then role-play as a future generation through various scenarios, they are expected to think and act in the best interests of the future. Taking inspiration from both these studies we adapted the ICECAP-A to include various scenarios and gave indicators to help put individuals in the 'role of the future generation'. **Results:** On-going. But I will be testing the validity of this new

expected well-being survey through focus groups and statistical analysis. **Relevance:** This research is relevant as it is important for the government to understand the effect policy changes will have on people's well-being before they implement them.

Name Company or University: TNO

(If applicable) External supervisor: Geerte Paradise

UU Supervisor: Samson Chota

Student Name: Tom Xavier Woods

Internship:

Title: What's the craic motivation?

Craicter is a rewards-based social platform, a start-up in the very early stages. To progress to further stages of accelerator programmes, secure investment, and succeed overall, understanding the customer is key. To do this, extensive surveys were carried out to validate the management team's assumptions and to understand what motivates potential users. 185 people were surveyed online and on the streets in Dublin. A mix of direct and indirect questioning was used to identify participants' daily habits and to get a feel of how they would use the app. The Leisure Motivation Scale and Big 5 Personality Test were applied to gain a deeper understanding of the potential users. It was found that 86% of people would use an app of this description and 50% would pay a monthly subscription of €5 per month. Participants that would use the app are intrinsically motivated to take part in leisure activities to accomplish something ($t = -2.357$, $df = 51$, $p = <0.05$) and to experience stimulation ($t = -2.394$, $df = 51$, $p = <0.05$). The company will tailor the design and functionality of the Craicter app to meet these needs.

Name of company: The Craicter Scale, or Craicter for short.

UU Supervisor: Sjoerd Stuit

Name Student: Carlotta Campisano

Internship or Thesis: Thesis

Title project: Where do You Keep Your Running Shoes?

In the Covid-19 crisis, people from all over the world were forced to spend a lot of time at home. For many, this meant having to come up with new strategies to create a home environment supportive of a healthy diet and exercise. This study aims to predict at-home health behavior by using a combination of nudge theory and the theory of planned behavior. Specifically, this study investigates whether three self-nudge types (visibility, accessibility, availability) and perceived control can be used as predictors of healthy eating and exercise. In this context, perceived control is defined as the estimation of how difficult or easy the execution of the behavior is going to be, based on both external resources and internal factors. To test the hypothesis of a covariate interaction between nudge and perceived control, an online 55-item survey regarding health habits and home arrangements was administered to 100 participants aged 17-73. The responses were analyzed using two multiple linear regressions, one for healthy eating and one for exercise. The results pointed towards a covariate interaction between nudge visibility and perceived

control for healthy eating but not exercise, where the only significant predictor was perceived control. These results suggest that the most effective strategy to implement healthy eating is a combination of heightened perceived control and an increase in the visibility of healthy eating self-nudges in the home environment.

Name company or University: Utrecht University

UU Supervisor: Krista Overvliet

Name Student: Naomi Sieben

Internship or Thesis: Thesis

Title project: The sound and visual perception of Urban Air Mobility vehicles

Urban Air Mobility (UAM) is a new popular concept for air vehicles, such as parcel delivery drones and air taxis, in an urban environment. Despite many positive aspects of this implementation, there are also negative sides, such as noise pollution. Little is known about the impact of drone noise and other types of UAM vehicles, but what research has shown is that annoyance rates towards drone noise is generally higher than that of existing conventional aircraft noise. A virtual reality experiment is conducted with 30 participants to measure the perception of UAM noise, with and without their visual appearance. Here, a busy urban area and a quieter urban area in Amsterdam were simulated using the Visual Community Noise Simulator (VCNS). In these virtual environments, annoyance scores were measured for drones, as well as helicopters and lawnmowers with and without a visual model. Additionally, two questionnaires were filled in on paper about pre-existing attitudes towards drones and general sound sensitivity. Results show however no differences in annoyance scores towards drones that are visually present, or only audible. Also no differences in annoyance scores to drones were found in the quieter and louder streets. Unexpectedly though, hovering drones were considered more annoying than the fly-over of a drone. Furthermore, a significant difference was found for vehicle type, where helicopters were rated as least annoying, followed by drones and lastly lawnmowers. This result was found for these vehicles with and without the visual appearance. Finally, only pre-existing attitudes towards drones were found as significant predictor of drone noise annoyance. These results are especially interesting since this is one of the first studies where the visual perception of drones is measured and compared to helicopters and lawnmowers. However, due to limitations in this study, such as testing participants at home because of Covid19-restrictions, future research is necessary to validate these results.

Name company or University: Netherlands Aerospace Centre (NLR)

External supervisor: Roalt Aalmoes

UU Supervisor: Jana Klaus

Name Student: Joep Franssen

Internship or Thesis: Thesis

Title project: Capturing the Customer's Mind: Comparison between Indirect and Direct Methods for Measuring Emotions throughout a Customer Journey

Abstract: Companies measure emotions throughout a customer journey to improve the user experience of products and services. To measure emotions, many companies use direct methods like the Net Promoter Score (NPS). In the present study, we map emotions throughout the sign-up process for a Greenwheels subscription using direct and indirect methods. We have set-up an indirect priming task in which subjects indicated whether a target word matches their emotions at different journey stages. By combining three key measures - choice, response latency, and choice consistency - we aimed to measure valence, arousal, and trust throughout the journey. We compared the outcomes of the priming task with the Net Promoter Score and explicit survey responses. The present study showed that the customer journey evoked stronger negative than positive emotions in the indirect priming task. Although we found more detractors than promoters resulting in a negative NPS score, we did not find a significant positive or negative correlation between the priming task and NPS score. The findings of the present study suggest that the NPS and explicit survey responses elicit more social desirable behavior than the indirect priming task.

Name company or University: Bloakes
(If applicable) External supervisor: Guus van Loon
UU Supervisor: Dr. Baptist Liefoghe

Name Student: Samiksha Gupta

Internship or Thesis: Thesis

Title project: Patient Experience in MRI: Eliminating the need for sedatives in pediatric patient scan; *A feasibility and user acceptance study towards a proposed intervention by identifying facilitators and barriers of similar interventions*

Background: There are several digital interventions aimed at pediatric learning, coaching, and behavior change. However, little is known on the right strategies to facilitate engagement and user acceptance of the digital interventions. There is still high dropout and low engagement rates witnessed for certain digital interventions. Owing to the complexity of integrating a digital intervention into the present and established healthcare workflow, sufficient research is needed on the correct strategies to develop, design, and implement a digital intervention. This study aims to identify facilitators and barriers towards user acceptance of digital interventions for children based on different stakeholders and users, whilst also outlining the necessary strategies towards acceptance and integration of a digital intervention. **Method:** This literature review study utilizes Google Scholar and PubMed databases to filter out the relevant research papers. The literature was searched based on relevant keywords specific to the research question. The repeated themes were then identified with the use of NVivo 12. A total of 56 articles were utilized in this paper.

Results: Based on the reviewed literature, three key stakeholders were identified: children, parents and health care professionals. The facilitators and barriers towards the user acceptance were then categorized on the basis of the stakeholders in the form of strategies and guidelines. Interactive and gamified content, personalization and customization, feedback, monitoring, social rewards and workflow streamlining were some of the key facilitators identified. Furthermore, developing and designing the intervention in a collaborative team of designers, child psychologist, doctors, and other stakeholders have

resulted positive user engagement. On the other hand, lack of language options, technical illiteracy and costs, cultural insensitivity and workflow disruptions were some of the key barriers towards the user acceptance of digital interventions. **Conclusion:** Behavior change interventions including education, training, and coaching with parent and healthcare professional involvement are effective in pediatric patient care. Efforts to expand the clinical workforce providing and assisting digital interventions and to educate the children, parents, and health care providers about the benefits of digital interventions can increase awareness and demand for these services. Broader calls by prominent professional organizations as well as health care professionals to help families replace physical discipline practices with more effective alternative disciplines can further elevate the issue of improved acceptance of digital interventions.

Name company: Philips

(If applicable) External supervisor: Sanne Nauts

UU Supervisor: David Terburg

Name Student: Clara Simon

Internship or Thesis: Thesis

Title project: Exploring the effect of cognitive variables on waiting time perception and satisfaction: OutSystems as a use case

In this study, we apply current findings of time perception to improve the loading design of the OutSystems platform to (1) increase user satisfaction, (2) tolerance for waiting and (3) reduce perceived time duration. The Attentional Gate Model (AGM) (1997) explains the internal processes of how humans estimate time. It is used to predict how long an event is perceived, based on two factors. By controlling those factors, it can therefore give suggestions on how to shorten time estimations and explain why certain moments “feel” longer than others. Based on the AGM, waiting design principles were applied to the design of a loading moment in the development process in OutSystems. The resulting designs were tested in a behavioral study. The findings of this study are not available, yet. The research contributes to the evolution of the OutSystems platform by giving a cognitive perspective on the perception of loading time.

Name company or University: Utrecht University/OutSystems

External supervisor: Tania Ramos

UU Supervisor: Martijn Mulder

Name student: Sarah Sheedy

Thesis or Internship: Thesis

Title Project: Factors Influencing Young Engineers’ Motivation to Join a Professional Association

The Royal Netherlands Society of Engineers, known as KIVI, has many members that are old and retiring. In order to support the many young engineers in the Netherlands, it is crucial for to understand the factors that influence the decision not to join KIVI. Such information could provide important data for strengthening organizational membership and expanding

representation within the profession. In addition, Nieuw Initiatief is working for KIVI on creating an online community to engage members. This thesis aimed to explore the underlying factors that inhibit young engineers from joining a networking association and highlight the most important aspects that members desire in an online community. In order to do so, small focus group interviews were conducted to initially understand how young engineers feel about professional associations, which inspired the creation of a questionnaire that was sent to a larger group of young engineers. Data analysis revealed that the lack of youth membership can be attributed to a lack of awareness, lack of perceived benefits and the associated costs. The most relevant findings were gathered and highlighted in separate reports for both Nieuw Initiatief and KIVI.

Name company or University: KIVI and Nieuw Initiatief

External Supervisor: Edwin de Kuiper and Gaby Wullink

UU Supervisor: Martijn Mulder

Name Student: Iris Pelsma

Internship or Thesis: Thesis

Title project: The effect of age, perceptual speed and numeracy on the performance of data visualizations

Introduction. In case of a gas leakage, the fire brigade and citizens need to understand the situation fast and accurately to make correct decisions, like to evacuate a building or to flee to a shelter. Gas cloud visualizations can help in correctly assessing the situation. Research has shown that cognitive abilities, as perceptual speed and numeracy influence the performance on data visualizations. **Methods.** In this study we investigated the influence of age, perceptual speed and numeracy on the performance of gas cloud visualizations (Cone and Tracks). **Results.** Results show that age has a negative effect on perceptual speed and numeracy. Besides, high scorers on perceptual speed showed shorter completion times for both visualizations. **Discussion.** This study confirms the theory that age can be used as an approximation for the cognitive abilities perceptual speed and numeracy. Moreover, high scorers on perceptual speed tend to have lower completion times. These people react faster in critical situations. The Cone and Tracks can both be used in case of a gas leakage.

Name company or University: TNO

(If applicable) External supervisor: Ward Venrooij

UU Supervisor: Yentl de Kloe